Generative-Model-Aided Grant-Free Random Access for GNSS-Independent MIMO-AFDM Satellite IoT Systems

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***Abstract*—This paper investigates grant-free random access in low Earth orbit satellite Internet-of-Things systems without global navigation satellite system (GNSS) assistance, where devices operate without time and frequency precompensation. We propose a multi-frame affine frequency division multiplexing (AFDM) transmission scheme that accommodates large differential delays and Doppler shifts across devices and analyze its input-output relationship. Additional pilot frames provide more observations without increasing the number of channel coefficients to be estimated. Based on this model, we learn a score-based generative channel prior directly from AFDM effective-channel samples, without imposing sparse or grid-aligned channel assumptions. From the learned prior, we construct a minimum mean-squared error (MMSE) channel denoiser that also quantifies estimation uncertainty for message passing. We then incorporate the denoiser into a message-passing (MP) receiver with two modules: multi-frame linear estimation (MFLE), which suppresses inter-user interference; and generative-prior-aided joint denoising (GPJD), which jointly updates channel and activity estimates. GPJD employs a density-ratio estimator that learns the likelihood ratio required for activity detection directly from samples, bypassing explicit evaluation of the marginal likelihood under the learned channel prior. The two modules iteratively exchange soft information for joint activity detection and channel estimation (JADCE). Numerical simulations demonstrate that the proposed receiver outperforms the baseline receiver in activity detection and channel estimation, while requiring fewer pilot frames to reach the same channel-estimation target.**

***Index Terms*—Affine frequency division multiplexing, grant-free random access, joint activity detection and channel estimation, score-based generative models.**

I. Introduction

Sixth-generation (6G) wireless networks are expected to provide ubiquitous connectivity for a massive number of Internet-of-Things (IoT) devices [1], [2]. To serve devices in remote areas with limited terrestrial coverage, low Earth orbit (LEO) satellites complement cellular networks for applications such as smart agriculture and climate monitoring [3], [4]. These power-limited devices transmit short packets sporadically, with only a small fraction active at a given time [1], [3], [5]. In grant-based random access, handshaking before transmission incurs significant signaling overhead, while the long propagation delay of terrestrial-satellite links further increases access latency [1], [2]. Accordingly, grant-free random access (GFRA) allows active devices to transmit pilots and data directly without first obtaining a grant, reducing signaling overhead and access latency in LEO satellite IoT systems [1], [2].

Supporting GFRA over LEO links requires handling large delays and Doppler shifts caused by long propagation distances and high satellite mobility. The 3rd Generation Partnership Project (3GPP) Release 17 framework addresses these effects through device-side precompensation assisted by the global navigation satellite system (GNSS) [6]. With such precompensation, many existing satellite GFRA schemes assume perfect compensation or small residual delays and Doppler shifts at the receiver [3]. Frequent estimation and compensation increase energy consumption and reduce battery lifetime, making these schemes less suitable for power-limited remote IoT devices [3]. This motivates reducing dependence on GNSS, consistent with the GNSS-independent operation considered in the Release 19 study on satellite access [7]. With reduced reliance on device-side precompensation, the satellite receiver must accommodate large differential delays and Doppler shifts, which can cause severe inter-symbol and inter-carrier interference in conventional orthogonal frequency division multiplexing (OFDM) systems [3], [8].

Orthogonal time frequency space (OTFS) modulation exploits the quasi-static properties of channels in the delay–Doppler domain to mitigate these effects [3], [9]. In [3], multi-frame OTFS transmission handles large differential delays and Doppler shifts at the satellite receiver, allowing devices to operate without GNSS assistance or precompensation of these offsets. Affine frequency division multiplexing (AFDM) offers a low-complexity modulation structure based on one-dimensional chirp signals [8], [10]. Its modulator can be implemented using a single inverse fast Fourier transform and two stages of pointwise chirp multiplication [11, Secs. II-B and III-A].

Accurate joint activity detection and channel estimation (JADCE) is essential for AFDM-GFRA [10]. Existing methods exploit sporadic device activity to formulate JADCE as a compressed sensing (CS) problem and incorporate structured priors to capture channel sparsity [3], [10], [12]. The scheme in [10] further uses a graph neural network to adaptively learn sparsity patterns, while retaining a predefined delay–Doppler grid for the sparse channel representation. In practical propagation environments, significant power leakage caused by fractional Doppler shifts and fractional time delays can weaken

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the sparsity of the effective channel matrix [8]. This motivates learning effective-channel priors from data to capture channel structures beyond the assumed sparse representation [1], [8].

Score-based generative models learn channel priors from samples by estimating the score function of the channel distribution [13], [14]. In [8], score-based learning is performed directly on the AFDM effective channel to address fractional delay–Doppler off-grid leakage in channel estimation. For massive random access, [15] proposes a two-stage algorithm that first identifies active devices through a Transformer-based detector and then uses generative diffusion models for joint channel estimation and data detection. Both approaches rely on iterative posterior sampling, which requires repeated score function evaluations and incurs substantial computational cost [8], [15].

The score-based turbo message passing (STMP) algorithm in [1] incorporates generative channel priors for JADCE without iterative posterior sampling. Through Tweedie's formulas, separate first- and second-order score networks provide posterior means and variances for single-step minimum mean-squared error (MMSE) denoising under equivalent additive white Gaussian noise (AWGN) observations. The relation between denoiser derivatives and estimation uncertainty [16] offers another way to obtain variance information: a trained denoiser can also quantify uncertainty without additional training [17]. This allows the learned channel prior to supply both channel estimates and variance information for message passing through the same denoiser. Moreover, STMP computes activity probabilities using Gaussian extrinsic messages obtained by projecting the channel posterior onto a Gaussian distribution [1]. For a non-Gaussian channel prior, matching the posterior mean and variance does not generally preserve the marginal likelihood. The relation between density ratios and class probabilities [18] offers an alternative: estimating the activity likelihood ratio directly from samples, without relying on this Gaussian approximation or explicitly evaluating the marginal likelihood.

Against this background, we investigate GNSS-independent AFDM-GFRA for LEO satellite IoT systems. Our main contributions are summarized as follows:

1) We propose a multi-frame AFDM transmission scheme that accommodates large differential delays and Doppler shifts at the satellite receiver without requiring device-side precompensation. We derive the input-output relationship and show that the effective channels across frames differ only by Doppler-induced phase rotations. Separating these phase rotations allows the received pilot frames to be jointly described by a shared effective channel matrix. Additional pilot frames therefore provide more observations without increasing the number of channel coefficients to be estimated. Compared with the multi-frame OTFS representation of Shen et al. [3], this effective-channel representation contains fewer coefficients and provides a compact model for learning the channel prior.
2) We learn a score-based generative prior for AFDM effective channels without imposing sparse or grid-aligned channel assumptions. Building on Tweedie's formula and the relation between denoiser derivatives and estimation uncertainty, we develop a channel denoiser that provides both channel estimates and average variance without requiring a separately trained variance model.
3) We develop a message-passing JADCE receiver comprising two modules: multi-frame linear estimation (MFLE), which jointly processes the pilot frames to suppress inter-user interference, and generative-prior-aided joint denoising (GPJD), which incorporates the proposed channel denoiser. Exploiting the relation between density ratios and class probabilities, we incorporate a learned density-ratio estimator into GPJD, enabling activity detection under the learned channel prior without explicit marginal-likelihood evaluation. The two modules iteratively exchange extrinsic messages to perform JADCE. Simulations show that the proposed receiver outperforms the STMP baseline adapted to AFDM in activity detection and channel estimation, and requires fewer pilot frames to achieve the same channel-estimation accuracy.

The remainder of this paper is organized as follows. Section II presents the system model and the multi-frame AFDM input–output relationship. Sections III–V describe the turbo message-passing framework, the learned channel denoiser and activity classifier, and the overall algorithm with its computational complexity analysis, respectively. Section VI presents the simulation results. Section VII concludes this paper.

Notation: Bold lowercase and uppercase letters denote vectors and matrices, respectively. $\mathcal{CN}(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ denotes a proper complex Gaussian distribution [19]. $\|\cdot\|_F$ denotes the Frobenius norm; $(\cdot)^{\mathsf{H}}$ and $\mathrm{tr}(\cdot)$ denote conjugate transpose and trace. $\mathrm{vec}(\cdot)$ stacks matrix columns, and $\mathbf{I}_n$ is the $n \times n$ identity matrix.

II. System Model

We consider a LEO satellite-enabled grant-free random access system as shown in Fig. 1, where AFDM is adopted to mitigate the effects of doubly dispersive terrestrial-satellite links (TSLs) [3], [8]. The system comprises K single-antenna terrestrial devices that communicate with a LEO satellite equipped with a uniform planar array (UPA) consisting of $M = M_y M_z$ antennas. During each transmission block, active devices share the same resources for pilot transmission to the satellite. These terrestrial devices operate without GNSS capabilities and do not pre-compensate for delay and Doppler shifts [3].

A. Terrestrial-Satellite Link Model

The TSLs experience fast variations due to the high mobility of the LEO satellite. Hence, this time variation and multipath propagation require a doubly dispersive channel model [3]. We assume that the terrestrial devices are stationary and all paths from the kth device have a common Doppler shift ν_k [8]. Within a transmission block, the path gains, delays, Doppler shifts, and angles of arrival remain approximately constant. We set $t_{\mathrm{c}} = 0$ at the beginning of the transmission block. After delay compensation for τ_{c} at the satellite receiver, the

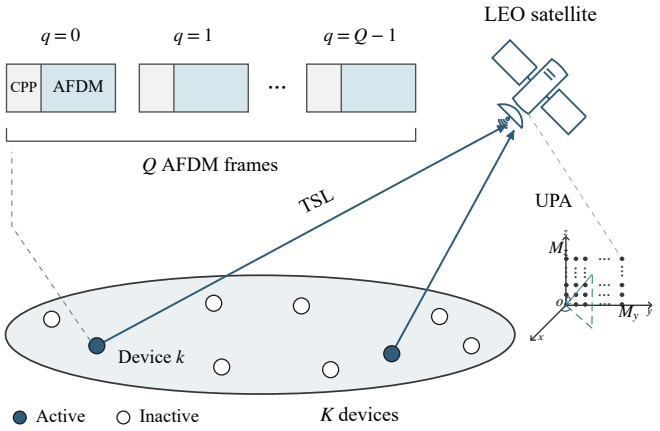


Fig. 1. LEO satellite grant-free access with multi-frame AFDM pilot transmission. The insets show the frame structure of active device k and the satellite UPA. Device activity is fixed over the Q frames. Satellite, footprint, and array artwork is adapted from [3, Fig. 1].

channel impulse response from the kth device to the satellite at time instant $t_{\rm c}$ can be represented as

$$\mathbf{h}_k(t_{\rm c},\tau)=\mathbf{a}_k\sum_{\ell=1}^{L_k}g_{k\ell}e^{{\rm j}2\pi\nu_k t_{\rm c}}\delta(\tau-\tau_{k\ell}),\tag{1}$$

where $k=1,\ldots,K$, L_k is the number of physical paths, $g_{k\ell}$ is the path gain of the ℓth path, $\delta(\cdot)$ is the Dirac delta function, and ${\rm j}^2=-1$. Here $\tau_{k\ell}=\widetilde{\tau}_{k\ell}-\tau_{\rm c}$ represents the differential delay for the ℓth path from the kth device to the satellite, where $\widetilde{\tau}_{k\ell}$ and $\tau_{\rm c}$ denote the propagation delay and common delay, respectively. The common delay corresponds to the propagation time between the satellite and the closest point on Earth within the beam coverage [3].

The array response vector $\mathbf{a}_k$ at the satellite side can be written as [3], [20]

$$[\mathbf{a}_k]_{n_z+M_z n_y}=e^{{\rm j}2\pi(d_y n_y\mu_{y,k}+d_z n_z\mu_{z,k})/\lambda},\tag{2}$$

where $n_z=0,\ldots,M_z-1$, $n_y=0,\ldots,M_y-1$, λ is the wavelength, and d_v is the spacing between adjacent antennas along the v-axis with $v\in\{y,z\}$. Here $\mu_{z,k}$ and $\mu_{y,k}$ are the directional cosines along the z-axis and y-axis, respectively.

B. Transmission Scheme

In the proposed scheme, each active device transmits pilot symbols over Q consecutive AFDM frames within one transmission block, sharing N chirp subcarriers with the other active devices [1], [8]. Each frame is modulated and demodulated independently, while the received frames are jointly processed for JADCE. Let $\mathbf{p}_{qk}\in\mathbb{C}^N$ denote the known pilot vector of device k in frame q, where $q=0,\ldots,Q-1$. The pilot vectors are collected in $\mathbf{P}_k=[\mathbf{p}_{0k},\ldots,\mathbf{p}_{Q-1,k}]^{\mathsf{T}}$ and $\mathbf{P}=[\mathbf{P}_1,\ldots,\mathbf{P}_K]\in\mathbb{C}^{Q\times KN}$, where $(\cdot)^{\mathsf{T}}$ denotes transpose. The pilots satisfy $\mathbf{P}\mathbf{P}^{\mathsf{H}}=KE_p\mathbf{I}_Q$, where $\|\mathbf{p}_{qk}\|^2=E_p$ for all k and q [1]. The construction of $\mathbf{P}$ is given in Section VI-A.

Applying the inverse discrete affine Fourier transform (IDAFT) to $\mathbf{p}_{qk}$ gives the time-domain transmitted signal [10]

$$s_{qk}[t]=\frac{1}{\sqrt{N}}\sum_{n=0}^{N-1}[\mathbf{p}_{qk}]_n e^{{\rm j}2\pi(c_1t^2+c_2n^2+nt/N)},\tag{3}$$

where $t=0,\ldots,N-1$, and c_1 and c_2 are AFDM modulation parameters. The signal in (3) can be represented in matrix form as [10]

$$\mathbf{s}_{qk}=\mathbf{F}_{\rm A}^{\mathsf{H}}\mathbf{p}_{qk},$$

where

$$\mathbf{F}_{\rm A}=\left[\frac{1}{\sqrt{N}}e^{-{\rm j}2\pi(c_2r^2+rt/N+c_1t^2)}\right]_{r,t=0}^{N-1}\tag{4}$$

denotes the discrete affine Fourier transform (DAFT) matrix. Here $T_s=1/(N\Delta f)$ gives the sampling interval, with Δf denoting the subcarrier spacing.

Assume that the chirp-periodic prefix (CPP) duration is greater than or equal to the maximum differential delay, i.e., $N_{\rm CPP}T_s\geq\max_{k,\ell}\tau_{k\ell}$. Each AFDM frame spans a duration of $T_{\rm frame}=(N+N_{\rm CPP})T_s$.

For fractional delays, we use the continuous-time AFDM waveform in [21, Sec. 3.1.1] with a chirp-periodic extension over the prefix [11]. The basis for chirp subcarrier n at time ξT_s is

$$\varphi_n(\xi)=\frac{1}{\sqrt{N}}e^{{\rm j}2\pi[c_2n^2+c_1\xi^2+nw_\xi/N-\zeta_n(w_\xi)w_\xi]},\tag{5}$$

where $w_\xi=\langle\xi\rangle_N$, $\langle x\rangle_N=x-N\lfloor x/N\rfloor$, $\zeta_n(w)=\lfloor 2c_1w+n/N\rfloor$, and $2Nc_1\in\mathbb{N}_+$.

After CPP removal and DAFT demodulation, the reference-frame channel from device k to antenna m can be expressed as

$$\mathbf{H}_{k,m}=[\mathbf{a}_k]_m\sum_{\ell=1}^{L_k}g_{k\ell}\mathbf{G}(\tau_{k\ell}/T_s,\nu_k;N_{\rm CPP}),\tag{6}$$

whose rows and columns correspond to received and transmitted symbols, respectively, with $m=0,\ldots,M-1$. Define $\mathbf{H}_k=[\mathbf{H}_{k,0}^{\mathsf{T}},\ldots,\mathbf{H}_{k,M-1}^{\mathsf{T}}]\in\mathbb{C}^{N\times MN}$. For a path with gain 1, delay dT_s, and Doppler shift ν, the channel response can be expressed as

$$\mathbf{G}(d,\nu;g)=\mathbf{F}_{\rm A}\mathbf{D}_{\nu,g}\mathbf{B}_d,\tag{7}$$

where $\mathbf{D}_{\nu,g}={\rm diag}\{e^{{\rm j}2\pi\nu T_s(g+t)}\}_{t=0}^{N-1}$ and $[\mathbf{B}_d]_{tn}=\varphi_n(t-d)$. Here $t_{\rm c}=(g+t)T_s$ for $t=0,\ldots,N-1$ after removing the CPP, and $g=N_{\rm CPP}$ for frame 0.

In frame q, the useful samples start at $g_q=N_{\rm CPP}+q(N+N_{\rm CPP})$. Thus, (7) gives

$$\mathbf{H}_k^{(q)}=e^{{\rm j}q\beta_k}\mathbf{H}_k,\tag{8}$$

where $\beta_k\equiv 2\pi\nu_kT_{\rm frame}\pmod{2\pi}$ and $\beta_k\in[-\pi,\pi)$. Since the paths of each device share a common Doppler shift, the inter-frame variation is described by the scalar phase rotation $e^{{\rm j}q\beta_k}$, while $\mathbf{H}_k$ remains approximately constant over Q consecutive frames. The received signals can therefore be expressed using this shared channel and the effective pilot matrix

$$\mathbf{A}_k={\rm diag}\{e^{{\rm j}q\beta_k}\}_{q=0}^{Q-1}\mathbf{P}_k,\tag{9}$$

where $\mathbf{A}=[\mathbf{A}_1,\ldots,\mathbf{A}_K]$.

In (6), the effective channel at each antenna is represented by an $N \times N$ matrix. The multi-frame OTFS scheme in [3, eq. (9)] uses an $L \times N \times N$ tensor for N subcarriers and L OFDM symbols per frame. The dense AFDM channel representation contains MN^2 channel entries per device, compared with MLN^2 in that OTFS representation. With the same subcarrier spacing and prefix length, L AFDM frames span the duration of one OTFS frame and contain the same number of useful samples.

C. Problem Formulation

We model device activities as independent and identically distributed (i.i.d.) binary variables. For device k, define α_k as [1]

$$\alpha_k = \begin{cases} 1, & \text{if the } k\text{th device is active,} \\ 0, & \text{otherwise.} \end{cases}$$

Let ρ represent the probability of a device being active, i.e., $\Pr(\alpha_k = 1) = \rho$, where $\rho \ll 1$. As a result, only a small subset of the devices is active in any given transmission block. The activity states are assumed to remain constant within a transmission block. Let $\mathbf{X}_k = \alpha_k \mathbf{H}_k$.

The received symbols in frame q after CPP removal and DAFT demodulation are collected in $\mathbf{Y}_q \in \mathbb{C}^{M \times N}$, with noise matrix $\mathbf{W}_q$ containing independent $\mathcal{CN}(0, \sigma^2)$ entries. Across the transmission block, the received frames form $\mathbf{Y} = [\mathrm{vec}(\mathbf{Y}_0^{\mathsf{T}}), \ldots, \mathrm{vec}(\mathbf{Y}_{Q-1}^{\mathsf{T}})]^{\mathsf{T}}$, with $\mathbf{W}$ arranged in the same way. With $\mathbf{X} = [\mathbf{X}_1^{\mathsf{T}}, \ldots, \mathbf{X}_K^{\mathsf{T}}]^{\mathsf{T}}$, the received signal is

$$\mathbf{Y} = \sum_{k=1}^{K} \mathbf{A}_k \mathbf{X}_k + \mathbf{W} = \mathbf{A}\mathbf{X} + \mathbf{W}, \tag{10}$$

where $\mathbf{Y}, \mathbf{W} \in \mathbb{C}^{Q \times MN}$, $\mathbf{A} \in \mathbb{C}^{Q \times KN}$, and $\mathbf{X} \in \mathbb{C}^{KN \times MN}$. This block formulation follows [1, Sec. II-A], with the device matrices defined by the AFDM channel in (6). Since the Q frames share the same $\mathbf{X}$, additional pilot frames provide more observations without increasing the number of channel coefficients to be estimated. The matrix $\mathbf{X}$ is block-sparse, with $\mathbf{X}_k = \mathbf{0}$ for inactive devices [1]. For $Q < KN$, the inverse problem in (10) is underdetermined.

Our goal is to jointly detect active devices and estimate their channels from the observation $\mathbf{Y}$ within a Bayesian inference framework [1]. Power leakage arises from fractional delays and Doppler shifts, which can weaken the sparsity of $\mathbf{H}_k$ [8]. We therefore estimate the full $\mathbf{H}_k$ using the training-channel distribution $q_{\mathrm{tr}}(\mathbf{H}_k)$ as the prior, without imposing channel sparsity or an element-wise factorization. We assume that $\mathbf{H}_k$ and α_k are independent for different values of k, and that $\mathbf{H}_k$ is independent of α_k. With $\widehat{\mathbf{A}}$ and $\widehat{\sigma}^2$ fixed as in Section V-B, these independence assumptions and Bayes' theorem yield the joint posterior factorization [1]

$$\begin{aligned} &q(\mathbf{X}, \{\mathbf{H}_k\}_{k=1}^K, \{\alpha_k\}_{k=1}^K \mid \mathbf{Y}) \\ &\quad \propto \prod_{j=1}^{MN} p(\mathbf{y}_j \mid \mathbf{x}_j) \\ &\quad \times \prod_{k=1}^{K} p(\mathbf{X}_k \mid \mathbf{H}_k, \alpha_k) q_{\mathrm{tr}}(\mathbf{H}_k) p(\alpha_k), \end{aligned} \tag{11}$$

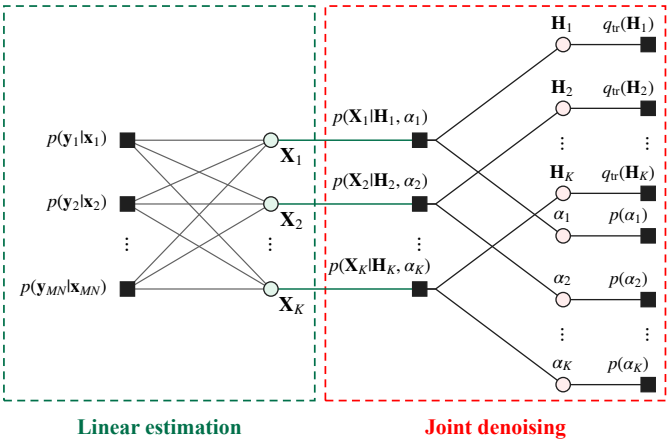


Fig. 2. Factor graph of (11) using device blocks $\mathbf{X}_k$ as variable nodes. Each likelihood factor depends on the corresponding column entries from all device blocks. Circles and squares denote variables and factors. The channel and activity prior branches are adapted from [1, Fig. 2].

where $\mathbf{x}_j$ and $\mathbf{y}_j$ denote the jth column of $\mathbf{X}$ and $\mathbf{Y}$, respectively. Here $p(\mathbf{y}_j \mid \mathbf{x}_j) = \mathcal{CN}(\mathbf{y}_j; \widehat{\mathbf{A}}\mathbf{x}_j, \widehat{\sigma}^2 \mathbf{I}_Q)$, $p(\mathbf{X}_k \mid \mathbf{H}_k, \alpha_k)$ enforces $\mathbf{X}_k = \alpha_k \mathbf{H}_k$, and $p(\alpha_k) = \rho^{\alpha_k}(1-\rho)^{1-\alpha_k}$. The inference targets are the MMSE channel estimate $\mathbb{E}_q[\mathbf{X}_k \mid \mathbf{Y}]$ and the posterior activity probability $q(\alpha_k = 1 \mid \mathbf{Y})$.

III. Turbo Message Passing Framework

Figure 2 shows the factor graph of the posterior in (11). Following the turbo message-passing framework in [1], the receiver alternates between a linear estimator that exploits the multi-frame likelihood and a joint denoiser that exploits the channel and activity priors. The two modules exchange Gaussian extrinsic messages based on expectation propagation [22].

A. Linear Estimation

The linear estimator receives a Gaussian message from joint denoising, with mean $\mathbf{X}^{\mathrm{pri}}$ and per-column covariance $\gamma \mathbf{I}_{KN}$, where $\gamma > 0$. Combining this message with the likelihood in (11) gives the posterior mean and covariance [1, Sec. III-A], [23, eq. (15c)]:

$$\mathbf{X}_{\mathrm{L}} = \mathbf{X}^{\mathrm{pri}} + \gamma \widehat{\mathbf{A}}^{\mathsf{H}} (\widehat{\sigma}^2 \mathbf{I}_Q + \gamma \widehat{\mathbf{A}} \widehat{\mathbf{A}}^{\mathsf{H}})^{-1} (\mathbf{Y} - \widehat{\mathbf{A}} \mathbf{X}^{\mathrm{pri}}), \tag{12}$$

$$\mathbf{\Sigma}_{\mathrm{L}} = (\gamma^{-1} \mathbf{I}_{KN} + \widehat{\sigma}^{-2} \widehat{\mathbf{A}}^{\mathsf{H}} \widehat{\mathbf{A}})^{-1}. \tag{13}$$

The posterior covariance is approximated by $t_{\mathrm{L}} \mathbf{I}_{KN}$, where $t_{\mathrm{L}} = \frac{\mathrm{tr}\, \mathbf{\Sigma}_{\mathrm{L}}}{KN}$ [1], [24].

The extrinsic message is obtained by dividing this Gaussian approximation by the incoming message [22]. Its variance and mean are

$$v_{\mathrm{L}} = (t_{\mathrm{L}}^{-1} - \gamma^{-1})^{-1}, \tag{14}$$

$$\mathbf{R}_{\mathrm{L}} = v_{\mathrm{L}} \left(\frac{\mathbf{X}_{\mathrm{L}}}{t_{\mathrm{L}}} - \frac{\mathbf{X}^{\mathrm{pri}}}{\gamma} \right). \tag{15}$$

We partition $\mathbf{R}_{\mathrm{L}}$ by device and pass the resulting matrix blocks to joint denoising with common variance v_{L}.

B. Joint Channel Estimation and Activity Detection

Joint denoising combines the channel estimate under the active hypothesis with the posterior activity probability to estimate $\mathbf{X}_k$ [1, Sec. III-B]. For each device, the incoming Gaussian message with mean $\mathbf{R}_k$ and variance v corresponds to the local observation model

$$\begin{aligned} \mathbf{R}_k &= \alpha_k \mathbf{H}_k + \mathbf{W}_k, \\ \mathrm{vec}(\mathbf{W}_k) &\sim \mathcal{CN}(\mathbf{0}, v\mathbf{I}_D), \end{aligned} \tag{16}$$

where $\mathbf{H}_k \sim q_{\mathrm{tr}}$ and $D = MN^2$.

1) Channel Estimation: Under the active hypothesis $\alpha_k = 1$, channel estimation reduces to denoising the AWGN observation in (16). The channel prior and Gaussian likelihood determine the conditional mean and average variance, which we approximate as

$$\mathbf{H}_k^{\mathrm{post}} \approx \mathbb{E}[\mathbf{H}_k \mid \mathbf{R}_k, \alpha_k = 1], \tag{17}$$

$$t_k \approx \frac{1}{D} \operatorname{tr} \mathrm{Cov}(\mathrm{vec}(\mathbf{H}_k) \mid \mathbf{R}_k, \alpha_k = 1). \tag{18}$$

Direct evaluation of these moments is difficult because it involves high-dimensional integrals over $\mathbf{H}_k$ under the channel prior q_{tr}.

2) Activity Detection: The inactive and active hypotheses, $\mathbf{R}_k = \mathbf{W}_k$ and $\mathbf{R}_k = \mathbf{H}_k + \mathbf{W}_k$, differ in their observation distributions. Under the active hypothesis, the likelihood $q_{1,v}$ averages over the unknown channel $\mathbf{H}$ with prior q_{tr}; under the inactive hypothesis, $q_{0,v}$ is the Gaussian noise density:

$$q_{1,v}(\mathbf{R}) = \mathbb{E}[\mathcal{CN}(\mathrm{vec}(\mathbf{R}); \mathrm{vec}(\mathbf{H}), v\mathbf{I}_D)], \tag{19}$$

$$q_{0,v}(\mathbf{R}) = \mathcal{CN}(\mathrm{vec}(\mathbf{R}); \mathbf{0}, v\mathbf{I}_D). \tag{20}$$

With prior activity probability ρ, Bayes' rule gives

$$p^\star(\mathbf{R}, v) = \frac{\rho q_{1,v}(\mathbf{R})}{(1-\rho) q_{0,v}(\mathbf{R}) + \rho q_{1,v}(\mathbf{R})}. \tag{21}$$

Computing $q_{1,v}$ requires high-dimensional integration over the unknown channel, making direct evaluation of this probability difficult. Let p_k denote the approximation to $p^\star(\mathbf{R}_k, v)$ used in the joint estimate below.

3) Joint Estimation and Extrinsic Message: Combining the activity probability p_k with a Gaussian approximation to the active-channel posterior, with mean $\mathbf{H}_k^{\mathrm{post}}$ and covariance $t_k \mathbf{I}_D$, gives a Bernoulli–Gaussian approximation to the posterior of $\mathbf{X}_k = \alpha_k \mathbf{H}_k$ [25, Secs. III-C–III-D], [1, Sec. III-B]. Its mean is

$$\widehat{\mathbf{X}}_k = p_k \mathbf{H}_k^{\mathrm{post}}. \tag{22}$$

Rearranging $\mathbf{H}_k^{\mathrm{post}}$ into $\mathbf{U}_k \in \mathbb{C}^{M \times N^2}$, with one row per antenna, gives the average posterior covariance across antennas

$$\mathbf{C}_{X,k} = p_k t_k \mathbf{I}_M + \frac{p_k(1-p_k)}{N^2} \mathbf{U}_k \mathbf{U}_k^{\mathsf{H}}. \tag{23}$$

This covariance accounts for both channel uncertainty under the active hypothesis and uncertainty in the activity state. Averaging over devices and antennas gives the posterior variance

$$\begin{aligned} \bar{t}_X &= \frac{1}{KM} \sum_k \operatorname{tr} \mathbf{C}_{X,k} \\ &= \frac{1}{K} \sum_k \left[p_k t_k + p_k(1-p_k) \frac{\left\| \mathbf{H}_k^{\mathrm{post}} \right\|_F^2}{D} \right]. \end{aligned} \tag{24}$$

The joint estimate is represented by a Gaussian approximation with mean $\widehat{\mathbf{X}}_k$ and common variance $\bar{t}_X$. As in (14)–(15), division by the incoming Gaussian message gives the extrinsic variance and mean, with $b = \bar{t}_X / v$:

$$\gamma_{\mathrm{out}} = \frac{vb}{1-b}, \tag{25}$$

$$\mathbf{X}_{k,\mathrm{out}} = \frac{\widehat{\mathbf{X}}_k - b\mathbf{R}_k}{1-b}. \tag{26}$$

For $0 < b < 1$, the extrinsic variance is positive. The output means form $\mathbf{X}^{\mathrm{pri}}$, and γ_{out} is the input variance of the linear estimator in the next iteration. Section IV addresses the computation of the conditional channel moments and activity probability required by these updates.

IV. Learning Channel Denoising and Activity Detection

This section develops learning methods for the conditional channel moments and activity probability required by Section III-B. The channel moments can be expressed through score functions of the noisy-channel distribution. We use denoising score matching to learn a channel denoiser and obtain the variance in (18) from its input derivatives [16], [17]. For activity detection, a classifier estimates the density ratio in (21).

A. Channel Mean Estimation

Under the active hypothesis in (16), the complex-valued channel $\mathbf{H} \sim q_{\mathrm{tr}}$ is corrupted by proper complex Gaussian noise with covariance $v\mathbf{I}_D$, where $v > 0$. The noise is independent of $\mathbf{H}$. Stacking the real and imaginary parts of $\mathrm{vec}(\mathbf{H})$ as $\mathbf{h}_{\mathbb{R}} \in \mathbb{R}^{2D}$ gives the real-valued observation model

$$\mathbf{u} = \mathbf{h}_{\mathbb{R}} + \sqrt{v/2}\, \boldsymbol{\epsilon}, \qquad \boldsymbol{\epsilon} \sim \mathcal{N}(\mathbf{0}, \mathbf{I}_{2D}). \tag{27}$$

At fixed v, let $q_v^{\mathbb{R}}$ denote the density of $\mathbf{u}$. The conditional mean is given by Tweedie's formula [16], [26]

$$\mathbf{m}_{\mathbb{R}}(\mathbf{u}, v) = \mathbf{u} + \frac{v}{2} \nabla_{\mathbf{u}} \log q_v^{\mathbb{R}}(\mathbf{u}), \tag{28}$$

where $\nabla_{\mathbf{u}} \log q_v^{\mathbb{R}}$ is the first-order score function [27].

Let $\mathbf{s}_\theta(\mathbf{u}, v)$ approximate the first-order score, with trainable parameters θ. Although the score of $q_v^{\mathbb{R}}$ is unknown, the conditional Gaussian density in (27) has score $-2(\mathbf{u} - \mathbf{h}_{\mathbb{R}})/v$. Denoising score matching [28] therefore uses

$$\mathcal{L}_{\mathrm{DSM}}(\theta) = \mathbb{E}\left[\frac{v}{4D} \left\| \mathbf{s}_\theta(\mathbf{u}, v) + \frac{2}{v}(\mathbf{u} - \mathbf{h}_{\mathbb{R}}) \right\|^2 \right]. \tag{29}$$

For each fixed v, this loss has the same minimizers as the mean-squared error between the network output and the score of the perturbed channel distribution.

To obtain the channel estimate directly, we parameterize the score as $\mathbf{s}_\theta = 2(\mathbf{f}_\theta - \mathbf{u})/v$, where $\mathbf{f}_\theta(\mathbf{u}, v)$ is a denoiser. Substituting this relation into (29) gives the training loss

$$\mathcal{L}_{\mathrm{ch}}(\theta) = \mathbb{E}\left[\frac{1}{Dv} \left\| \mathbf{f}_\theta(\mathbf{u}, v) - \mathbf{h}_{\mathbb{R}} \right\|^2 \right]. \tag{30}$$

The expectation is taken over the clean channels, noise variances, and Gaussian noise. For each fixed v, the conditional

mean $\mathbf{m}_{\mathbb{R}}(\mathbf{u}, v)$ minimizes this loss over all estimators of $\mathbf{h}_{\mathbb{R}}$ from $\mathbf{u}$.

The denoiser is trained over a range of noise variances [1, Sec. IV-B] to accommodate the input messages in (16). Its real and imaginary outputs are reshaped into the complex-valued estimate $\mathbf{F}_\theta(\mathbf{R}, v)$. The sampling distribution of v is specified in Section VI-A2, and the architecture and preconditioning of $\mathbf{f}_\theta$ are given in Section VI-A2.

B. Channel Variance Estimation

The conditional covariance $\mathbf{P}_{\mathbb{R}} = \mathrm{Cov}(\mathbf{h}_{\mathbb{R}} \mid \mathbf{u}, v)$ is related to the second-order score [16], [1, Sec. IV-A]:

$$\mathbf{P}_{\mathbb{R}} = \frac{v}{2}\mathbf{I}_{2D} + \frac{v^2}{4}\nabla^2_{\mathbf{u}} \log q_v^{\mathbb{R}}(\mathbf{u}). \tag{31}$$

Here the second-order score is the Hessian of the log density.

Differentiating (28) with respect to $\mathbf{u}$ at fixed v and comparing the result with (31) relates the covariance to the mean Jacobian [16, Prop. 1]. Since the joint denoiser uses the average variance in (18), only the covariance trace is needed:

$$\mathbf{P}_{\mathbb{R}} = \frac{v}{2}\mathbf{J}_m, \tag{32}$$

$$\tau^\star = \frac{v}{2D}\,\mathrm{div}\,\mathbf{m}_{\mathbb{R}}. \tag{33}$$

Here $\mathbf{J}_m = \partial \mathbf{m}_{\mathbb{R}}/\partial \mathbf{u}$ is the Jacobian at fixed v, and $\tau^\star$ is the average conditional variance per complex coordinate. Thus, the average variance can be estimated from the derivatives of the trained denoiser without training a separate variance model [17, Sec. 3.2 and Sec. 4.1].

For numerical evaluation, we approximate the divergence by Monte Carlo derivative estimation [29], using central finite differences of $\mathbf{f}_\theta$ to estimate $\tau^\star$. Let $\mathbf{z}_\ell$ be independent random sign vectors with $\mathbb{E}[\mathbf{z}_\ell \mathbf{z}_\ell^{\mathsf{T}}] = \mathbf{I}_{2D}$, and define $\mathbf{w}_\ell = \sqrt{v/2}\,\mathbf{z}_\ell$. Using L_z random directions and a finite-difference step $\delta_{\mathrm{FD}} > 0$, the resulting estimate is

$$\widehat{\tau}_\theta(\mathbf{R}, v) = \frac{1}{2\delta_{\mathrm{FD}} D L_z} \sum_{\ell=1}^{L_z} \mathbf{w}_\ell^{\mathsf{T}} \Delta_\ell \mathbf{f}_\theta, \tag{34}$$

where $\Delta_\ell \mathbf{f}_\theta = \mathbf{f}_\theta(\mathbf{u} + \delta_{\mathrm{FD}}\mathbf{w}_\ell, v) - \mathbf{f}_\theta(\mathbf{u} - \delta_{\mathrm{FD}}\mathbf{w}_\ell, v)$. The finite-difference settings are given in Section VI-A4.

C. Activity Probability Estimation

The activity probability depends on the likelihoods through their ratio, as seen by writing (21) in log-odds form:

$$\log \frac{p^\star(\mathbf{R}, v)}{1 - p^\star(\mathbf{R}, v)} = \log \frac{\rho}{1-\rho} + \log \frac{q_{1,v}(\mathbf{R})}{q_{0,v}(\mathbf{R})}. \tag{35}$$

Direct evaluation of $q_{1,v}$ requires the channel integral in (19). Binary classification instead allows the ratio $q_{1,v}/q_{0,v}$ to be estimated from observations drawn under the two hypotheses [18].

The two hypotheses define the training classes $a \in \{0, 1\}$, with equal prior probability $\Pr(a = 1) = 1/2$. The active class $a = 1$ consists of observations $\mathbf{R} = \mathbf{H} + \mathbf{W}$ with $\mathbf{H} \sim q_{\mathrm{tr}}$, whereas the inactive class $a = 0$ contains noise-only observations $\mathbf{R} = \mathbf{W}$. The noise satisfies $\mathrm{vec}(\mathbf{W}) \sim \mathcal{CN}(\mathbf{0}, v\mathbf{I}_D)$ independently of $\mathbf{H}$, and both classes use the same sampling distribution of v. We train the classifier with parameters ϕ by minimizing the binary cross-entropy loss

$$\mathcal{L}_{\mathrm{act}}(\phi) = \mathbb{E}[\log(1 + e^{z_\phi}) - a z_\phi], \tag{36}$$

where $z_\phi = z_\phi(\mathbf{R}, v, \mathbf{F}_\theta(\mathbf{R}, v))$. For this balanced training distribution, the minimizer of the expected loss over all functions of $(\mathbf{R}, v)$ is the log likelihood ratio

$$z^\star(\mathbf{R}, v) = \log \frac{q_{1,v}(\mathbf{R})}{q_{0,v}(\mathbf{R})}. \tag{37}$$

The classifier output therefore estimates the log density ratio [18, Sec. 4.2, eq. (17)]. For $\mathbf{F}_k = \mathbf{F}_\theta(\mathbf{R}_k, v)$ and $\ell_k = z_\phi(\mathbf{R}_k, v, \mathbf{F}_k)$, combining this estimate with the receiver's activity prior in (35) gives

$$p_k = \mathrm{sigmoid}(\mathrm{logit}\,\rho + \ell_k), \tag{38}$$

where $\mathrm{logit}\,a = \log[a/(1-a)]$ and $\mathrm{sigmoid}\,b = (1 + e^{-b})^{-1}$.

V. GP-MP Algorithm

A. Overall Algorithm

The proposed generative-prior message-passing receiver, denoted by GP-MP, combines MFLE in Section III-A with GPJD. The learned denoiser and activity classifier in Section IV implement the channel and activity updates within GPJD, as illustrated in Fig. 3.

For an input Gaussian message with mean $\mathbf{R}_k$ and variance v, the denoiser provides the channel mean $\mathbf{F}_k = \mathbf{F}_\theta(\mathbf{R}_k, v)$ and the derivative-based variance estimate $\widehat{\tau}_k = \widehat{\tau}_\theta(\mathbf{R}_k, v)$ in (34). The common variance τ^{post} is the average of $\widehat{\tau}_k$ over k, subject to the positive lower bound $\tau_{\min}$. The classifier provides the corresponding activity probability p_k through (38).

Within GPJD, the channel prior is approximated by a Gaussian site $\mathcal{CN}(\mathrm{vec}(\mathbf{H}_k); \mathrm{vec}(\mathbf{E}_k), \tau_k^{\mathrm{ext}}\mathbf{I}_D)$ [30, Sec. 2.1]. Its parameters follow from dividing the Gaussian posterior approximation, with mean $\mathbf{F}_k$ and variance τ^{post}, by the input likelihood:

$$\tau_k^{\mathrm{ext}} = ((\tau^{\mathrm{post}})^{-1} - v^{-1})^{-1}, \tag{39}$$

$$\mathbf{E}_k = \tau_k^{\mathrm{ext}} \left(\frac{\mathbf{F}_k}{\tau^{\mathrm{post}}} - \frac{\mathbf{R}_k}{v} \right). \tag{40}$$

The product of this site and the current Gaussian likelihood determines the channel variance and mean under the active hypothesis:

$$t_k = \frac{\tau_k^{\mathrm{ext}} v}{\tau_k^{\mathrm{ext}} + v}, \tag{41}$$

$$\mathbf{H}_k^{\mathrm{post}} = \frac{\tau_k^{\mathrm{ext}} \mathbf{R}_k + v\mathbf{E}_k}{\tau_k^{\mathrm{ext}} + v}. \tag{42}$$

Together with p_k, these moments determine the joint channel estimate and the Gaussian extrinsic message returned to MFLE. Algorithm 1 summarizes the resulting iterations. The receiver uses phases that are either available or estimated once before MP by the method in Section V-B; they remain fixed throughout the iterations. Initialization uses the normalized training-channel power $\widehat{P}_H$, defined as the average of $\|\mathbf{H}\|_F^2/D$ over the training channels. Stopping checks and numerical settings are given in Section VI-A.

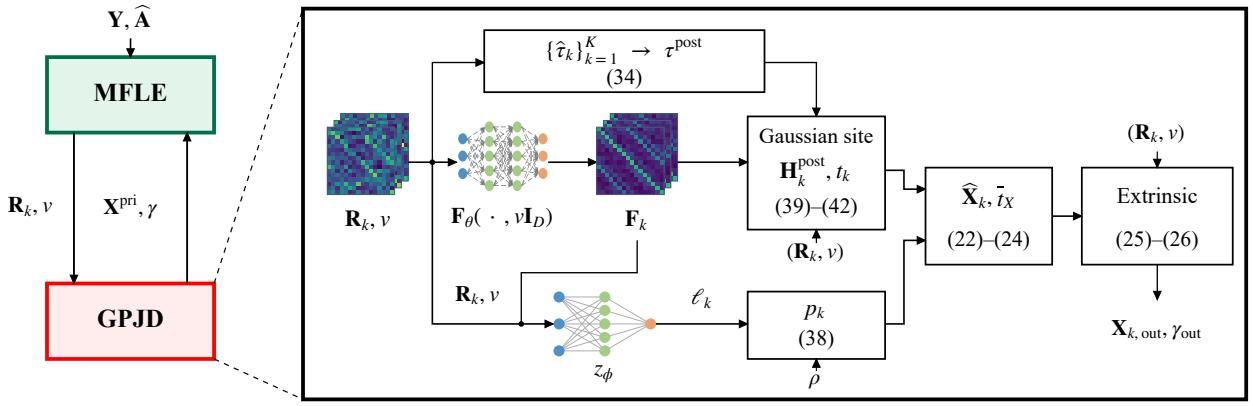


Fig. 3. Message exchange between MFLE and GPJD, with an expanded view of GPJD. Following [1, Fig. 3], the denoising illustration uses an input–output pair with a common magnitude scale.

Algorithm 1 GP-MP for JADCE

Input: $\mathbf{Y}$, $\{\mathbf{p}_{qk}\}$, available phases, ρ, trained $\mathbf{f}_\theta$ and z_ϕ.
Output: $\{s\widehat{\mathbf{X}}_k, s\mathbf{H}_k^{\mathrm{post}}, p_k, \widehat{\alpha}_k\}_{k=1}^K$.

Estimate $\widehat{\sigma}^2$; estimate unavailable phases once as in Section V-B.
Construct $\widehat{\mathbf{A}}$ using fixed phases and normalize as in Section V-B.
Compute the SVD of $\widehat{\mathbf{A}}$; draw one random sign vector per device and reuse it across iterations.
Initialize $\mathbf{X}^{\mathrm{pri},(0)} \leftarrow \mathbf{0}$, $\gamma^{(0)} \leftarrow \rho\widehat{P}_H$.
Initialize $\mathbf{E}_k \leftarrow \mathbf{0}$, $\tau_k^{\mathrm{ext}} \leftarrow 1$, $\kappa^{(1)} \leftarrow 1$.
for $i = 1, \ldots, I_{\max}$ **do**
  % MFLE
  Update $\mathbf{R}_{\mathrm{L},k}$ and v_{L} by (12)–(15), using (53)–(55).
  $\mathbf{R}_k \leftarrow \mathbf{R}_{\mathrm{L},k}$, $\bar{v} \leftarrow v_{\mathrm{L}}$; apply (49)–(50) for $i \geq 2$.
  $v \leftarrow \kappa^{(i)}\bar{v}$.
  % GPJD
  $\mathbf{F}_k \leftarrow \mathbf{F}_\theta(\mathbf{R}_k, v)$.
  Compute $\widehat{\tau}_k$ and p_k by (34) and (38).
  Average $\widehat{\tau}_k$ over k to obtain τ^{post}.
  Apply the lower bound $\tau_{\min}$ to τ^{post}.
  if $\tau^{\mathrm{post}} < v$ **then**
    Update $\mathbf{E}_k$ and τ_k^{ext} by (39)–(40).
  end if
  Update $\mathbf{H}_k^{\mathrm{post}}$ and t_k by (41)–(42).
  Update $\widehat{\mathbf{X}}_k$, $\mathbf{C}_{X,k}$, and $\bar{t}_X$ by (22), (23), and (24).
  Update $\kappa^{(i+1)}$ as in Section V-B.
  Compute $\mathbf{X}_{k,\mathrm{out}}$ and γ_{out} by (25)–(26) with clipping.
  $\mathbf{X}_k^{\mathrm{pri},(i)} \leftarrow \mathbf{X}_{k,\mathrm{out}}$, $\gamma^{(i)} \leftarrow \gamma_{\mathrm{out}}$; apply (51)–(52) for $i \geq 2$.
  Update the stopping history as in Section VI-A4.
  if the stopping conditions in Section VI-A4 hold **then**
    break
  end if
end for
$\widehat{\alpha}_k \leftarrow \mathbf{1}\{p_k \geq 1/2\}$.
return $\{s\widehat{\mathbf{X}}_k, s\mathbf{H}_k^{\mathrm{post}}, p_k, \widehat{\alpha}_k\}_{k=1}^K$.

B. Parameter Estimation and Numerical Implementation

Noise-variance estimation exploits the spatial covariance of the received signals. Let $\mathbf{y}_{qr}$ be column r of $\mathbf{Y}_q$ and $N_{\mathrm{s}} = QN > M \geq 2$. The spatial sample covariance and noise-variance estimate are

$$\widehat{\mathbf{R}}_y = \frac{1}{N_{\mathrm{s}}} \sum_{q,r} \mathbf{y}_{qr}\mathbf{y}_{qr}^{\mathsf{H}}, \tag{43}$$

$$\widehat{\sigma}^2 = \frac{\sum_{i=\widehat{r}_{\mathrm{s}}+1}^{M} \lambda_i}{M - \widehat{r}_{\mathrm{s}}}. \tag{44}$$

Here $\lambda_1 \geq \cdots \geq \lambda_M$ are the covariance eigenvalues, and $\widehat{r}_{\mathrm{s}}$ is selected by the Wax–Kailath minimum description length (MDL) criterion [31], [32]. Under a low-rank signal covariance and spatially white noise, the remaining eigenvalues provide the noise-variance estimate, which is held fixed during MP.

For phase estimation, the transposed frame observations are collected in $\mathbf{Y}_{\mathrm{s}} = [\mathbf{Y}_0, \ldots, \mathbf{Y}_{Q-1}]^{\mathsf{T}} \in \mathbb{C}^{QN \times M}$. Approximating each device's channel by one dominant path and absorbing the common scalar phase into the gain gives the response

$$[\mathbf{h}_k^{\mathrm{p}}(d,\nu)]_{qN+r} = e^{\mathrm{j}2\pi\nu q T_{\mathrm{frame}}} [\mathbf{G}_0(d,\nu)\mathbf{p}_{qk}]_r, \tag{45}$$

where $\mathbf{G}_0(d,\nu) = e^{-\mathrm{j}2\pi c_1 d^2}\mathbf{G}(d,\nu;0)$.

With $\mathbf{h}_k^{\mathrm{p}} = \mathbf{h}_k^{\mathrm{p}}(d_k,\nu_k)$, the observation model becomes $\mathbf{Y}_{\mathrm{s}} \approx \sum_k \mathbf{h}_k^{\mathrm{p}}\mathbf{b}_k^{\mathsf{T}} + \mathbf{W}_{\mathrm{s}}$, where the independent gains follow $\mathbf{b}_k \sim \mathcal{CN}(\mathbf{0}, \gamma_k\mathbf{I}_M)$, $\gamma_k \geq 0$. Integrating out the gains gives the Gaussian marginal-likelihood objective [33, Sec. 2.1]

$$\mathbf{S} = \widehat{\sigma}^2\mathbf{I}_{QN} + \sum_k \gamma_k \mathbf{h}_k^{\mathrm{p}}\mathbf{h}_k^{\mathrm{p},\mathsf{H}}, \tag{46}$$

$$\mathcal{L} = \log\det\mathbf{S} + \frac{1}{M}\operatorname{tr}(\mathbf{Y}_{\mathrm{s}}^{\mathsf{H}}\mathbf{S}^{-1}\mathbf{Y}_{\mathrm{s}}). \tag{47}$$

We minimize $\mathcal{L}$ by coordinate updates, solving analytically for each γ_k and refining (d_k,ν_k) using L-BFGS-B [34]. The search settings are given in Section VI-A4.

The estimated Doppler determines $\widehat{\beta}_k$ and $\widehat{\mathbf{A}}$ through (8)–(9). GP-MP uses the resulting phases throughout its iterations, with channel denoising based on the learned prior in Section IV.

Minimizing the expected Gaussian negative log likelihood over the input variance v, with the current approximate posterior in Section III-B3 held fixed [35, Sec. III-A], gives

$$v_{\mathrm{tar}} = \frac{1}{KD}\sum_{k=1}^{K}\left\|\mathbf{R}_k - \widehat{\mathbf{X}}_k\right\|_F^2 + \bar{t}_X. \tag{48}$$

This target adjusts the variance supplied to GPJD through $v^{(i)} = \kappa^{(i)}\bar{v}^{(i)}$ at iteration i, where $\bar{v}^{(i)}$ is the damped MFLE variance. The next scaling factor is a weighted average of

$\kappa^{(i)}$ and the ratio of $v_{\mathrm{tar}}^{(i)}$ to $\bar{v}^{(i)}$, with weights $1-\eta$ and η, respectively, where $\eta \in (0,1]$ is the damping factor.

The channel and observation quantities $(\mathbf{H}_k, \mathbf{X}_k, \mathbf{Y}, \mathbf{W})$ are normalized by the root mean square s of the training-channel entries, with noise variance scaled by s^2. The same symbols denote the normalized quantities.

Recursive damping [36, Sec. III-C] combines the current MFLE output with the previous input to GPJD. For $i \geq 2$,

$$\mathbf{R}_k^{(i)} = (1-\eta)\mathbf{R}_k^{(i-1)} + \eta \mathbf{R}_{\mathrm{L},k}^{(i)}, \tag{49}$$

$$\bar{v}^{(i)} = (1-\eta)\bar{v}^{(i-1)} + \eta v_{\mathrm{L}}^{(i)}. \tag{50}$$

The feedback updates in (25)–(26) use $b = \bar{t}_X / v$ clipped to $[\varepsilon, 1-\varepsilon]$, where $0 < \varepsilon < 1/2$, and are damped with the same factor:

$$\mathbf{X}_k^{\mathrm{pri},(i)} = (1-\eta)\mathbf{X}_k^{\mathrm{pri},(i-1)} + \eta \mathbf{X}_{k,\mathrm{out}}^{(i)}, \tag{51}$$

$$\gamma^{(i)} = (1-\eta)\gamma^{(i-1)} + \eta \gamma_{\mathrm{out}}^{(i)}. \tag{52}$$

The first input and feedback messages are used without damping.

C. Computational Complexity Analysis

With precomputed chirp factors, L AFDM frames require at most $LC_{\mathrm{FFT}}(N) + 2LN$ complex multiplications [11, eq. (16)], where $C_{\mathrm{FFT}}(n)$ is the cost of an n-point fast Fourier transform (FFT) or inverse FFT. For the same LN useful symbols, rectangular-pulse OTFS without transmit windowing requires $NC_{\mathrm{FFT}}(L)$ after cancelling the consecutive discrete Fourier transform (DFT) and inverse DFT operations [37].

For JADCE with fixed phases and $Q \leq KN$, the singular value decomposition (SVD), $\widehat{\mathbf{A}} = \mathbf{U}\,\mathrm{diag}(s_1, \ldots, s_Q)\mathbf{V}^{\mathsf{H}}$, is reused across iterations, where s_a are the singular values. The scalar updates in (13)–(14) can be written as

$$f_{\mathrm{L}} = \frac{1}{KN} \sum_{a=1}^{Q} \frac{\gamma s_a^2}{\widehat{\sigma}^2 + \gamma s_a^2}, \tag{53}$$

$$t_{\mathrm{L}} = \gamma(1 - f_{\mathrm{L}}), \tag{54}$$

$$v_{\mathrm{L}} = \frac{t_{\mathrm{L}}}{f_{\mathrm{L}}}. \tag{55}$$

Here $\mathbf{R}_{\mathrm{L}} = \mathbf{X}^{\mathrm{pri}} + (\mathbf{X}_{\mathrm{L}} - \mathbf{X}^{\mathrm{pri}})/f_{\mathrm{L}}$.

The cost per block comprises preprocessing and iterative estimation. Preprocessing includes the SVD and noise-variance estimation, with costs $O(KNQ^2)$ and $O(QNM^2 + M^3)$, respectively, as well as phase estimation in Section V-B, whose cost is denoted by C_{phase}. Per iteration, MFLE transformations cost $O(QKMN^2)$, spatial covariance operations cost $O(KM^2N^2)$, and matrix factorizations cost $O(M^3)$. For per-device denoiser and classifier costs C_θ, C_ϕ, each GPJD update in Algorithm 1 adds $K[(1+2L_z)C_\theta + C_\phi]$ to the per-iteration cost. This corresponds to $1 + 2L_z$ denoiser evaluations and one classifier evaluation per device. Denoting the total per-iteration cost by C_{iter}, the online complexity for I iterations is $O(C_{\mathrm{phase}} + QNM^2 + M^3 + KNQ^2 + IC_{\mathrm{iter}})$. Here $C_{\mathrm{phase}} = 0$ for known phases. The pilot matrix, SVD factors, channel messages, and spatial covariances require $O(QKN + KMN^2 + KM^2)$ memory.

TABLE I
SIMULATION PARAMETERS

Parameter	Value
Carrier frequency f_c	2 GHz
Satellite altitude	600 km
Beam radius	128 km
Minimum elevation angle	30°
UPA dimensions $M_y \times M_z$	4×4
Array element spacing d_y, d_z	$\lambda/2$
Number of chirp subcarriers N	16
Subcarrier spacing Δf	30 kHz
Chirp parameters (c_1, c_2)	$(7/32, 1/320)$
CPP length N_{CPP}	336
Differential-delay range	$[0, 699]$ μs
Doppler-shift range	$[-41, 41]$ kHz
Power delay profile	NTN-TDL-D
Delay spread	$[30, 50]$ ns
Rician factor	11.707 dB
Default (K, Q, p)	$(40, 16, 0.1)$

VI. SIMULATION RESULTS

Working-draft data status: GP-MP uses the white-noise-trained networks in Figs. 4–6 and Table III; GP-MP-EP in Fig. 6 uses the same networks. The GP-MP results in Figs. 7–10 use earlier checkpoints with different training perturbations and await replacement.

A. Simulation Settings

1) Channel Dataset and System Parameters: Table I summarizes the default settings. Devices are uniformly distributed over the beam footprint, with delays, Doppler shifts, and array responses determined by the geometry. Large-scale effects are compensated. We use the NTN-TDL-D profile [3], with a delay spread drawn uniformly from $[30, 50]$ ns, a line-of-sight gain of fixed amplitude and uniform phase, and proper complex Gaussian diffuse gains. Path parameters remain fixed per block. The training and validation sets contain 50,000 and 5,000 clean $16 \times 16 \times 16$ channels, respectively, drawn from disjoint sets of scenes. Each scene provides four channel samples before multiplication by α_k.

For $Q \leq KN$, the pilots are constructed from distinct DFT rows $r_q \in \{0, \ldots, KN-1\}$, with a column permutation $\{c_{kn}\}_{k,n}$ of the same set and independent phases $\phi_{kn} \sim \mathcal{U}[0, 2\pi)$:

$$[\mathbf{p}_{qk}]_n = \sqrt{\frac{E_p}{N}} e^{-\mathrm{j}2\pi r_q c_{kn}/(KN)} e^{\mathrm{j}\phi_{kn}}, \tag{56}$$

with $\sum_{n=0}^{N-1} |[\mathbf{p}_{qk}]_n|^2 = E_p$.

Independent Bernoulli activity variables with nominal probability p are conditioned on at least one active device. The receiver prior uses the resulting marginal activity probability $\rho = p/[1-(1-p)^K]$. The received SNR is the ratio of the average noiseless signal power per complex observation to the noise variance. The signal power is averaged over the QMN entries of $\mathbf{AX}$ in each scene, and σ^2 is chosen to achieve the specified SNR.

TABLE II
NETWORK ARCHITECTURES

Layer or stage	Channels	Map size
Channel denoiser		
Input convolution	128	$N \times N$
Residual stage 1	128	$N \times N$
Residual stage 2	256	$N/2 \times N/2$
Residual stage 3 (dilation 2)	256	$N/2 \times N/2$
Residual stage 4 (dilation 4)	256	$N/2 \times N/2$
Refinement blocks 1–2	$256, 256$	$N/2 \times N/2$
Refinement block 3	128	$N/2 \times N/2$
Refinement block 4	128	$N \times N$
Output convolution	$2M$	$N \times N$
Activity classifier		
Convolutional stage 1	64	$N \times N$
Convolutional stage 2	128	$N/2 \times N/2$
Convolutional stage 3	128	$N/4 \times N/4$
Fully connected layers	$128, 1$	—

2) Network Architectures and Training: The denoiser uses the RefineNet [38] variant in [13, Appendix A], with $2M$ real and imaginary input maps over $N \times N$ DAFT coordinates. Each stage contains two residual blocks, each with two 3×3 convolutions, conditional instance normalization, and exponential linear unit (ELU) activations. The stage outputs are combined by four refinement blocks that restore the spatial resolution, with 3×3 convolutions at the network input and output. Table II lists the channel widths and spatial resolutions of both networks.

Noise conditioning uses an embedding of $\log(v)/5$, formed by two 64-unit fully connected layers with sigmoid-weighted linear unit (SiLU) activations and linearly projected to the normalization layers. The denoiser also adopts the preconditioning in [39, Sec. 5]:

$$\begin{aligned} \mathbf{f}_\theta(\mathbf{u}, v) &= c_{\mathrm{skip}}(v)\mathbf{u} + c_{\mathrm{out}}(v)\mathbf{g}_\theta(c_{\mathrm{in}}(v)\mathbf{u}, v), \\ c_{\mathrm{in}}(v) &= (\sigma_{\mathrm{d}}^2 + v/2)^{-1/2}, \\ c_{\mathrm{skip}}(v) &= \frac{\sigma_{\mathrm{d}}^2}{\sigma_{\mathrm{d}}^2 + v/2}, \\ c_{\mathrm{out}}(v) &= \left(\frac{\sigma_{\mathrm{d}}^2 v/2}{\sigma_{\mathrm{d}}^2 + v/2}\right)^{1/2}. \end{aligned} \tag{57}$$

Here $\sigma_{\mathrm{d}}^2 = 1/2$ is the nominal variance per real coordinate after normalization, and $\mathbf{g}_\theta$ is the RefineNet described above.

The classifier estimates the log likelihood ratio from $4M$ real and imaginary maps of $\mathbf{R}$ and $\mathbf{F}_\theta(\mathbf{R}, v)$, scaled by $\sqrt{2/(1+v)}$, together with five scalars: $\log(v)/5$, logarithmic transforms of input/output/residual energies normalized by Dv, and a sign-preserving logarithmic transform of the real inner product normalized by Dv. Its three convolutional stages each contain two 3×3 convolutions with SiLU activations and padding 1. Downsampling uses stride 2 in the first convolution of stages 2–3; the remaining convolutions use stride 1. Noise conditioning is provided by a 64-unit fully connected SiLU embedding of $\log(v)/5$, whose linear projections are added to the stage outputs. The flattened features and five scalars are combined by the fully connected layers in Table II, with SiLU between them, to produce one logit per device.

Training minimizes (30) with the preconditioning in (57), using v sampled uniformly on a logarithmic scale over $[10^{-3}, 10^{2}]$. The denoiser is optimized using Adam for 200 epochs with batch size 64, initial learning rate 10^{-4}, and exponential moving average decay 0.999. Model selection uses the lowest validation loss under fixed noise realizations. The STMP reference uses the same clean channels and number of epochs.

With the denoiser fixed, the classifier is trained from random initialization by minimizing (36) for 100 epochs, with batch size 32 and initial learning rate 3×10^{-4}. The classifier is also selected by validation loss. Classifier training and receiver evaluation use white Gaussian noise with covariance $v\mathbf{I}$.

3) Compared Receivers: We evaluate GP-MP and five baselines. All use known phases except for GP-MP-EP:

- **GP-MP:** The proposed receiver; EP denotes estimated phases.
- **STMP:** We adapt STMP [1] to the multi-frame AFDM model, retaining its white-noise-trained first- and second-order score networks for conditional mean and variance estimation. We replace its Gaussian-message-based activity-probability update with a learned density-ratio estimator trained using the corresponding first-order network.
- **Simultaneous orthogonal matching pursuit (SOMP):** [40] This greedy method sequentially selects dictionary atoms shared by the receive antennas and estimates their coefficients by least squares. We use an integer delay–Doppler dictionary and select up to 64 atoms.
- **GAMP-SBL:** This method uses generalized approximate message passing in sparse Bayesian learning [41]. Our implementation uses the integer delay–Doppler dictionary, shares row precisions across receive antennas, and allows up to 1,000 iterations.
- **GNN-ASPL:** This method alternates AMP-based channel estimation with adaptive sparsity pattern learning using a graph neural network [10]. It uses the same integer delay–Doppler dictionary and a 10-layer network.
- **Off-grid SBL:** In addition to the channel coefficients, this method estimates continuous delay and Doppler offsets through a first-order linear approximation of the dictionary [42]. We allow up to 120 iterations. For the SNR comparison, the 7, 13, and 25 Doppler grid points are paired with 12, 24, and 48 retained candidates per device and 48, 96, and 192 offset candidates per iteration, respectively; all three configurations use 112 delay grid points.

4) Evaluation Settings and Metrics: Phase estimation in GP-MP-EP covers $d \in [0, N)$ and $|\nu| \leq \nu_{\max}$, where $\nu_{\max}$ is the channel model's Doppler bound. Pilot correlations over 128 phase bins identify three coarse candidates. The Doppler values corresponding to each candidate phase within this bound form the initial search set, from which the best 12 are refined by at most 25 L-BFGS-B iterations per candidate within a differentiable interval of the waveform. Coordinate refinement starts from zero powers and uses two sweeps in

TABLE III
ACTIVITY DETECTION PERFORMANCE AND ITERATION-LIMIT RATES

	STMP				GP-MP			
	Gaussian		Density ratio		Gaussian		Density ratio	
SNR (dB)	AER (%)	Limit (%)	AER (%)	Limit (%)	AER (%)	Limit (%)	AER (%)	Limit (%)
−4	9.225	2	0.525	0	44.575	63	0.200	0
−2	7.875	3	0.225	0	43.475	55	0.125	0
0	6.375	2	0.100	0	40.975	46	0.075	0
2	5.550	1	0.050	0	36.750	31	0.025	0
4	4.800	1	0.025	0	32.050	19	0.025	0
6	4.225	0	0.025	0	27.425	11	0.025	0

descending order of initial pilot-correlation scores, with the current parameters retained among the candidates.

All receivers estimate the noise variance from the observations. STMP and GP-MP use activity threshold 0.5 and default damping factors of 0.2 and 0.4, respectively. Their iterations are bounded by $I_{\min} = 60$ and $I_{\max} = 600$, with relative-change tolerance 5×10^{-4}. After $I_{\min}$ iterations, convergence is assessed from the stability of channel and activity estimates and the consistency of variance updates. Runs that reach $I_{\max}$ return the last iterate.

GP-MP uses $L_z = 1$ and $\delta_{\mathrm{FD}} = 10^{-2}$ in (34), with the random sign vectors reused across iterations. The common variance has a lower bound $\tau_{\min} = 10^{-4}$.

Receiver comparisons use paired scenes, with the same scenes reused across conditions. Each SNR point uses 1,000 scenes, whereas each point in the pilot-frame, device-count, activity-probability, and cross-profile figures uses 500 scenes. The activity-update comparison in Table III uses 100 paired scenes per SNR.

For scene indices $e \in \mathcal{T}$ at a given condition, the normalized mean-squared error (NMSE) is the sum of squared Frobenius errors between $\widehat{\mathbf{X}}_{ek}$ and $\mathbf{X}_{ek}$, divided by the sum of squared Frobenius norms of $\mathbf{X}_{ek}$. Both sums cover all scenes in $\mathcal{T}$ and all K devices, including inactive devices; the ratio is reported in decibels. For direct channel denoising, $\mathbf{X}$ is replaced by $\mathbf{H}$ and the sums run over individual channel samples. The activity error rate (AER) is the total number of missed detections and false alarms divided by the $K|\mathcal{T}|$ device decisions.

B. Channel Denoising Performance

Following [1, Sec. VI-B], we first assess channel denoising under AWGN using $\mathbf{R} = \mathbf{H} + \mathbf{W}$, with $\mathrm{vec}(\mathbf{W}) \sim \mathcal{CN}(\mathbf{0}, v\mathbf{I})$, known v, and nominal SNR $-10 \log_{10} v$. At each of six SNRs from -20 to 30 dB, both neural estimators use the same 2,000 samples from 500 scenes and the same white-noise realizations. The following reference methods are included:

- **Least squares (LS):** The estimate is the noisy observation $\mathbf{R}$.
- **Gaussian-prior estimate:** The estimate is $\widehat{P}_H \mathbf{R}/(\widehat{P}_H + v)$, where $\widehat{P}_H$ is the training-channel power defined in Section V-A.

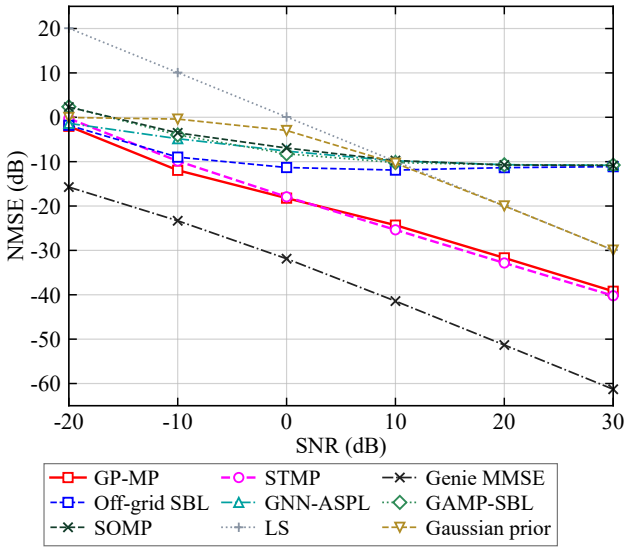


Fig. 4. Channel denoising under AWGN with known variance.

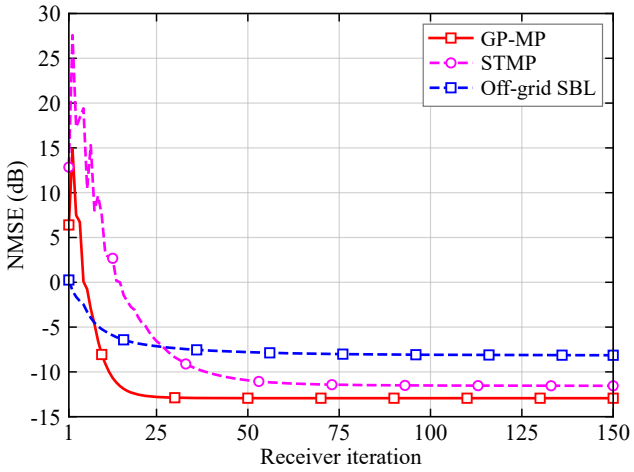


Fig. 5. NMSE versus iteration over 30 paired scenes at 0 dB, with $K = 40$, $Q = 16$, and $p = 0.1$.

- **Genie-aided conditional MMSE:** We compute the analytic conditional MSE of the remaining Gaussian gains given the true path geometry, powers, and line-of-sight gain.

The four sparse-recovery methods in Section VI-A3 are also evaluated using an identity observation operator and known AWGN variance. They retain their receiver settings and GNN weights; GAMP-SBL retains its internal noise-variance update.

In Fig. 4, both learned denoisers outperform LS and the Gaussian-prior estimate across the evaluated range. Their NMSE continues to decrease with SNR, whereas the four sparse-recovery methods approach error floors, leaving a wider gap at moderate and high SNR. The GP-MP denoiser has lower NMSE at low SNR, whereas the STMP denoiser is more accurate at high SNR. Both learned denoisers have higher NMSE than the genie-aided reference, which uses true path

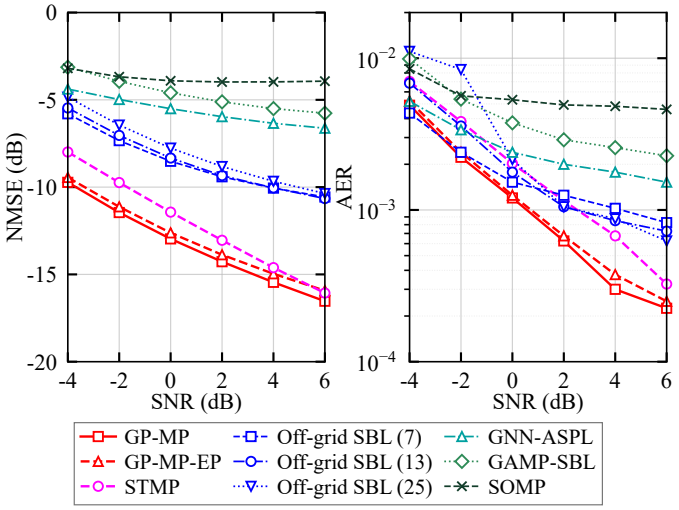


Fig. 6. Channel NMSE and AER versus SNR with $K = 40$, $Q = 16$, and $p = 0.1$, using 1,000 paired scenes per point. Numbers in the Off-grid SBL labels indicate the Doppler grid points.

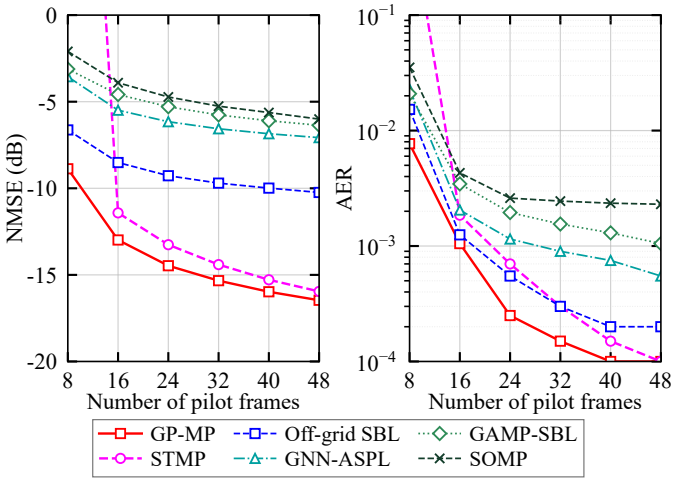


Fig. 7. Channel NMSE and AER versus the number of pilot frames at 0 dB, with $K = 40$ and $p = 0.1$.

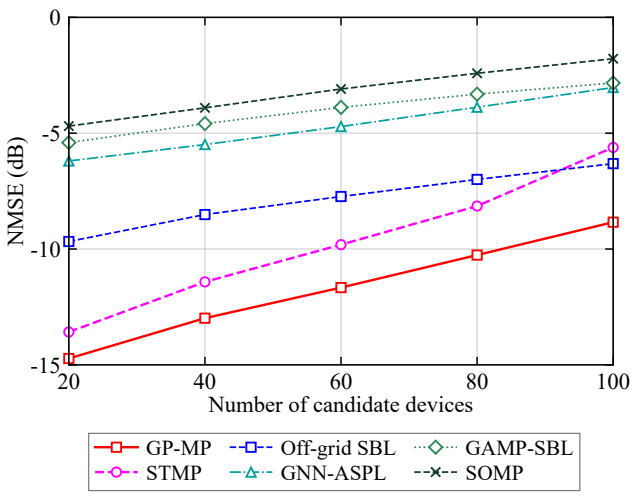


Fig. 8. Channel NMSE versus the number of candidate devices at 0 dB, with $Q = 16$ and $p = 0.1$.

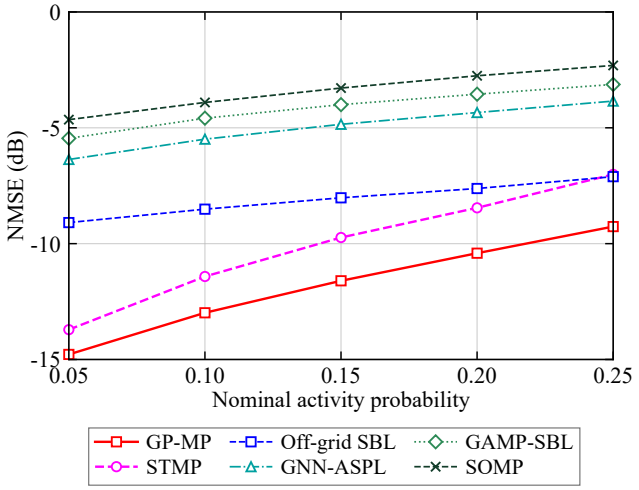


Fig. 9. Channel NMSE versus nominal activity probability at 0 dB, with $K = 40$ and $Q = 16$.

information.

C. Effect of Density Ratio Estimation

The effect of density ratio estimation is evaluated by comparing the learned activity update with the Gaussian approximation, for which

$$\ell_k^{\mathrm{G}} = \frac{\|\mathbf{R}_k\|_F^2}{v} - \frac{\|\mathbf{R}_k - \mathbf{E}_k\|_F^2}{\tau_k^{\mathrm{ext}} + v} - D \log\left(1 + \frac{\tau_k^{\mathrm{ext}}}{v}\right), \tag{58}$$

$$p_k = \mathrm{sigmoid}(\mathrm{logit}\, \rho + \ell_k^{\mathrm{G}}). \tag{59}$$

Each receiver, STMP and GP-MP, is evaluated with both activity updates under otherwise identical settings. The two GP-MP variants share the white-noise-trained denoiser, with the matched classifier used for density ratio estimation. Table III reports AER over all runs and the percentage reaching the 600-iteration limit (Limit).

For both receivers, the learned density ratio gives lower AER at every evaluated SNR, with no runs reaching the iteration limit. With the Gaussian approximation, STMP has lower AER than GP-MP throughout the evaluated range and reaches the iteration limit less frequently. The comparison therefore supports the use of learned density ratios in both receivers, with a larger improvement for GP-MP.

D. JADCE Performance

Unless stated otherwise, the receivers use the defaults in Table I and the evaluation settings in Section VI-A4.

Figure 5 compares the NMSE trajectories of GP-MP, STMP, and Off-grid SBL over 30 paired scenes, with early stopping disabled. GP-MP approaches a lower NMSE in fewer itera-

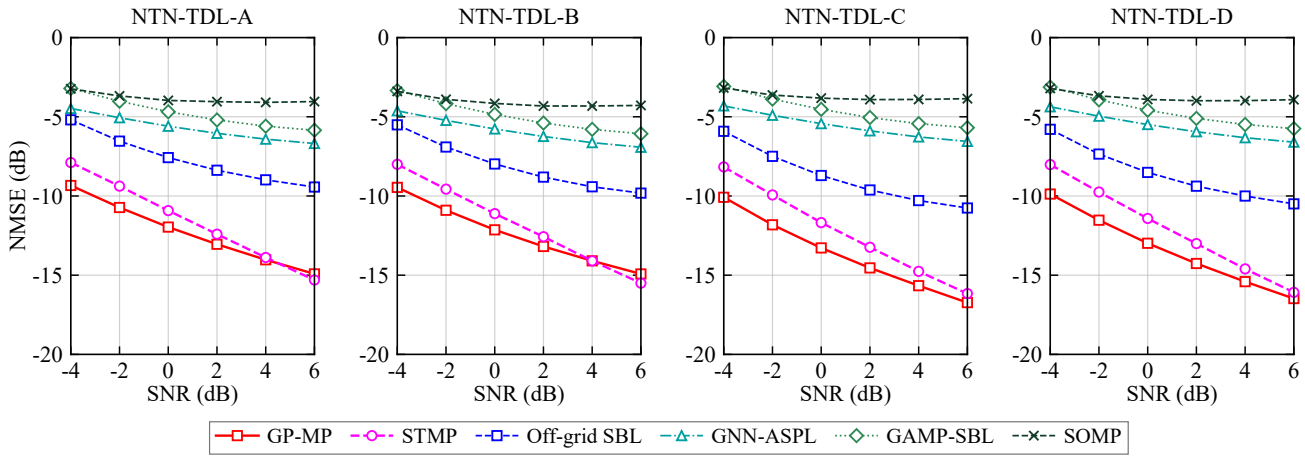


Fig. 10. Channel NMSE for NTN-TDL profiles A, B, C, and D (from left to right). Each point uses 500 paired physical scenes, $K = 40$, $Q = 16$, and $p = 0.1$. The learned receivers are trained on profile D and evaluated without retraining.

tions than STMP, while Off-grid SBL levels off at a higher NMSE.

Figure 6 compares channel NMSE and AER over -4 to 6 dB. GP-MP achieves the lowest NMSE throughout this range. Its advantage over STMP is more pronounced at low SNR and narrows as SNR increases. At 0 dB, GP-MP achieves an NMSE approximately 1.5 dB lower than STMP. Off-grid SBL has the lowest NMSE among the four sparse-recovery baselines. The NMSE of these baselines decreases more slowly with SNR than that of the learned receivers, so the gap between the two groups grows.

The AER also generally decreases with SNR. GP-MP has lower AER than STMP throughout the range and outperforms the four sparse-recovery baselines at moderate and high SNR.

With estimated phases, GP-MP-EP incurs an NMSE increase of 0.28–0.60 dB relative to GP-MP over the evaluated SNR range, while its AER remains close to the known-phase reference.

Figure 7 shows the effect of the number of pilot frames. Additional frames provide further observations of the same effective channels at fixed per-frame energy, and generally improve both channel estimation and activity detection. Among the evaluated frame counts, GP-MP first reaches -14 dB NMSE at $Q = 24$, whereas STMP requires $Q = 32$. This corresponds to 25% fewer pilot frames for the same NMSE target.

Figure 8 compares the channel NMSE for $K = 20$–100. At fixed activity probability and pilot-frame count, increasing K enlarges the set of candidate channels and raises the expected number of active devices. All methods exhibit higher NMSE as this load increases. GP-MP maintains the lowest NMSE, and its advantage over STMP grows with the number of candidate devices.

Figure 9 examines the effect of increasing the nominal activity probability from 0.05 to 0.25 with a fixed candidate set. With fixed pilot and antenna resources, more active channels must be recovered, and all methods exhibit higher NMSE. GP-MP retains the lowest NMSE, and its advantage over STMP increases with activity probability.

E. Comparison Across Channel Profiles

Performance beyond the training distribution is evaluated on NTN-TDL profiles A, B, and C using the networks trained on profile D without retraining. These evaluations vary the power delay profiles and fading distributions while retaining the geometry, pilots, and receiver settings, with 500 paired scenes per condition.

Figure 10 shows that GP-MP achieves lower NMSE than the four sparse-recovery baselines on every evaluated profile. This result indicates that the learned effective-channel prior remains useful beyond the training profile. The difference relative to STMP is smaller and varies with the profile and SNR.

VII. Conclusion

In this paper, we investigated GNSS-independent GFRA for LEO satellite IoT systems. We proposed a multi-frame AFDM transmission scheme that accommodates large differential delays and Doppler shifts without device-side precompensation. Under a shared effective-channel representation, additional pilot frames provide more observations without increasing the number of channel coefficients to be estimated. We learned a score-based generative prior for AFDM effective channels without imposing sparse or grid-aligned channel assumptions, and developed a denoiser that provides channel estimates and average variance without requiring a separately trained variance model. We incorporated this denoiser into a message-passing JADCE receiver comprising MFLE and GPJD. The learned density-ratio estimator in GPJD enables activity detection under the learned prior without explicit marginal-likelihood evaluation. Simulation results demonstrated improved activity detection and channel estimation over the adapted STMP baseline, with fewer pilot frames required for comparable channel-estimation performance. The receiver also achieved lower NMSE than the sparse-recovery baselines across the tested NTN-TDL profiles without retraining.

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